APP DEVELOPMENT

Build Mobile Apps. Learn Modern Technologies. Develop Real-World Applications.

Aayush Institute of Professional Studies offers career-oriented **App Development training** for students, graduates, freshers, working professionals and aspiring developers who want to learn how mobile applications are designed, developed, tested and prepared for deployment.

The programme introduces learners to mobile application development concepts, user interfaces, programming, databases, APIs, application testing and practical project development.

Depending on the selected programme, learners can explore **Android Development, Flutter Development and React Native Development**.

Our learning approach combines **technical concepts, hands-on practice, project development, industry-oriented tools and career preparation**.

WHAT IS APP DEVELOPMENT?

App Development is the process of designing, developing, testing and maintaining applications for mobile devices.

A modern mobile application can involve:

User Interface → Programming → Application Logic → Database → APIs → Authentication → Testing → Deployment

Mobile applications can be developed for different platforms using native or cross-platform technologies.

MOBILE APP DEVELOPMENT TECHNOLOGIES

ANDROID DEVELOPMENT

Build applications specifically for the Android ecosystem.

Technologies May Include:

- Android Studio
- Kotlin
- Java Fundamentals
- Android SDK

- XML / Modern Android UI Concepts
- Activities & Application Components
- Layouts & UI Components
- Navigation
- Data Storage
- API Integration
- Application Testing
- Debugging

FLUTTER APP DEVELOPMENT

Flutter is a cross-platform development framework that can be used to build applications for multiple platforms from a common codebase.

Topics May Include:

- Flutter Fundamentals
- Dart Programming
- Widgets
- Stateless & Stateful Widgets
- Layouts
- Navigation
- Forms
- User Input
- State Management Fundamentals
- API Integration
- JSON
- Local Data Storage
- Firebase Fundamentals

- Debugging
- Testing
- Application Build & Deployment

REACT NATIVE DEVELOPMENT

React Native enables developers to build mobile applications using React and JavaScript / TypeScript technologies.

Topics May Include:

- React Fundamentals
- React Native
- JavaScript / TypeScript
- Components
- Props & State
- Hooks
- Navigation
- Forms
- API Integration
- JSON
- Local Storage
- Authentication
- Application Debugging
- Build & Deployment Fundamentals

WHO CAN LEARN APP DEVELOPMENT?

App Development can be suitable for:

- 12th Pass Students

- ITI Students
- Polytechnic / Diploma Students
- BCA Students
- B.Sc. IT / Computer Science Students
- B.Tech / B.E. Students
- MCA Students
- M.Tech Students
- Computer Science / IT Graduates
- Freshers
- Working Professionals
- Existing Web Developers
- Freelancers
- Career Changers

Beginners can start with foundation-level programming before progressing to advanced application development.

COMPLETE APP DEVELOPMENT LEARNING PATH

01. COMPUTER & DEVELOPMENT FUNDAMENTALS

Build the foundation required for application development.

Topics Include:

- Computer Fundamentals
- Operating System Basics
- Internet Fundamentals
- Software Development Concepts
- Development Environment
- Files & Folders

- Command Line Fundamentals
- Basic Problem Solving
- Programming Concepts

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02. PROGRAMMING FUNDAMENTALS

Develop programming logic before building applications.

Topics May Include:

- Variables
- Data Types
- Operators
- Conditions
- Loops
- Functions
- Arrays
- Objects
- Classes & Objects
- Basic OOP Concepts
- Error Handling
- Problem Solving
- Debugging

Programming language depends on the selected technology track.

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03. MOBILE UI DEVELOPMENT

Learn how mobile application interfaces are designed and structured.

Topics Include:

- Mobile UI Fundamentals
- Screen Layouts
- Navigation
- Buttons
- Forms
- Input Fields
- Lists
- Cards
- Images
- Icons
- Menus
- Dialogues
- Responsive Layouts
- User Interaction

The objective is to create functional and user-friendly mobile interfaces.

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04. ANDROID DEVELOPMENT

For learners selecting the Android pathway.

Topics May Include:

- Android Studio
- Android Project Structure
- Android SDK
- Kotlin / Java Fundamentals
- Activities
- Application Components

- Layouts
- UI Components
- Navigation
- Permissions
- Local Data
- API Integration
- Notifications
- Debugging
- Testing

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05. DART & FLUTTER

For learners selecting the Flutter pathway.

Dart Fundamentals

- Variables
- Data Types
- Functions
- Collections
- Classes
- OOP Concepts
- Asynchronous Programming
- Error Handling

Flutter Development

- Flutter Architecture
- Widgets
- Layouts

- Stateful & Stateless Widgets
- Navigation
- Forms
- User Input
- State Management
- API Integration
- Local Storage
- Firebase Fundamentals
- Debugging
- Testing

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06. REACT NATIVE

For learners selecting the React Native pathway.

Topics May Include:

- React Fundamentals
- Components
- JSX
- Props
- State
- Hooks
- Navigation
- Forms
- API Integration
- JavaScript / TypeScript
- Local Storage

- Authentication
- Reusable Components
- Debugging

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07. DATABASE & LOCAL STORAGE

Mobile applications often need to store and retrieve information.

Topics May Include:

- Database Fundamentals
- Local Storage
- SQLite Fundamentals
- Data Management
- CRUD Operations
- Cloud Database Concepts
- Firebase Fundamentals
- Data Synchronisation

Technology coverage depends on the selected course.

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08. API & BACKEND INTEGRATION

Learn how mobile applications communicate with servers and external services.

Topics Include:

- API Fundamentals
- REST APIs
- HTTP Methods
- JSON
- GET / POST / PUT / PATCH / DELETE

- API Requests & Responses
- Authentication
- API Integration
- Error Handling
- Third-Party API Integration

Examples may include connecting an app with user accounts, products, orders, bookings, payments or other online services.

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09. USER AUTHENTICATION

Learn the fundamentals of secure user access.

Topics May Include:

- Registration
- Login
- Logout
- User Authentication
- Authorization
- Password Security Fundamentals
- Token-Based Authentication
- Session Concepts
- Access Control
- Basic Application Security

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10. FIREBASE & CLOUD SERVICES

Depending on the selected programme, learners may be introduced to Firebase and related cloud services.

Topics May Include:

- Firebase Fundamentals
- Authentication
- Cloud Database
- Cloud Storage
- Push Notifications
- Application Analytics
- Backend Service Integration

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11. APP TESTING & DEBUGGING

Learn how to identify and resolve application problems.

Topics Include:

- Application Testing
- UI Testing Fundamentals
- Functional Testing
- Debugging
- Error Identification
- Device Testing
- Emulator Testing
- API Testing
- Performance Fundamentals
- Bug Fixing

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12. GIT & VERSION CONTROL

Learn professional development practices for managing application source code.

Topics Include:

- Git Fundamentals
- Repository Management
- Commit
- Branch
- Merge
- GitHub
- Version Control
- Project Collaboration

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13. APP BUILD & DEPLOYMENT

Understand the basic process of preparing applications for release.

Topics May Include:

- Application Build Process
- Debug & Release Builds
- Application Configuration
- App Signing Fundamentals
- Version Management
- Deployment Concepts
- Google Play Store Fundamentals
- Application Distribution
- Maintenance & Updates

Actual store publication may require developer accounts, platform policies and technical requirements outside the training institute.

PRACTICAL APP DEVELOPMENT PROJECTS

Project work allows learners to apply programming and application development concepts.

BEGINNER PROJECTS

- Calculator App
- To-Do App
- Notes App
- Digital Profile App
- Simple Utility App
- Quiz Application

INTERMEDIATE PROJECTS

- Student Management App
- Attendance App
- Restaurant Menu App
- Appointment Booking App
- Expense Tracker
- News / Content Application
- Product Catalogue App

ADVANCED PROJECTS

- E-Commerce Mobile App
- Food Ordering App
- Service Booking App
- Education / Learning App
- Business Management App
- CRM Mobile Application
- Delivery Tracking Application
- API-Based Business Application

Project scope and technology depend on the selected programme and learner level.

SKILLS YOU CAN DEVELOP

Depending on the selected programme, learners can develop skills in:

- Mobile Application Development
- Android Development
- Kotlin / Java Fundamentals
- Flutter
- Dart
- React Native
- JavaScript / TypeScript
- Mobile UI Development
- API Integration
- Database Management
- Firebase
- Authentication
- Git & GitHub
- Application Testing
- Debugging
- App Deployment
- Problem Solving
- Project Development

NATIVE VS CROSS-PLATFORM APP DEVELOPMENT

NATIVE APP DEVELOPMENT

Native development focuses on building applications specifically for a particular platform.

Android → Kotlin / Java → Android SDK

This approach provides platform-specific development capabilities.

CROSS-PLATFORM APP DEVELOPMENT

Cross-platform technologies allow developers to build applications for multiple platforms using a shared development approach.

Popular Technologies:

Flutter → Dart

React Native → JavaScript / TypeScript

The appropriate technology depends on the project, platform requirements, development team and business objectives.

APP DEVELOPMENT CAREER PATHWAYS

App Development skills can support entry-level opportunities such as:

- Android Developer
- Mobile App Developer
- Flutter Developer
- React Native Developer
- Junior Mobile Developer
- Application Developer
- Mobile UI Developer
- Software Developer
- App Development Intern
- Mobile Application Intern

Actual job titles, eligibility, responsibilities and technology requirements vary by employer and role.

FREELANCING & APP PROJECT WORK

App Development skills can also support independent project work.

Potential project areas include:

- Business Applications
- E-Commerce Apps
- Booking Applications
- Educational Apps
- Service Applications
- Utility Apps
- Customer Applications
- Internal Business Applications
- Mobile UI Development
- Application Maintenance

Freelancing income and project availability depend on individual skills, portfolio, client requirements and market conditions.

LIVE PROJECT & PRACTICAL EXPOSURE

Depending on the selected programme, learners may receive opportunities for:

- Live Project Training
- Practical Application Development
- Mentor-Guided Projects
- API Integration
- Database Integration
- Testing & Debugging
- Project Documentation
- Application Presentation

- Portfolio Development
- Industry-Oriented Development Practices

The objective is to provide practical exposure to the application development lifecycle.

INTERNSHIP OPPORTUNITIES

Eligible learners may receive information about available internship opportunities in areas such as:

- Android Development
- Flutter Development
- React Native
- Mobile Application Development
- Software Development
- Application Testing

Paid internships may be available with certain organisations, subject to eligibility, availability and the company's selection process.

Stipend or compensation, where applicable, is determined by the organisation offering the internship.

CAREER GUIDANCE & PLACEMENT SUPPORT

Aayush Institute provides **career guidance at no additional cost** to help learners understand suitable technology pathways based on their qualification, interests, skills and career objectives.

Career support may include:

- Career Assessment
- Course Selection Guidance
- Resume / CV Preparation
- Portfolio Guidance
- Technical Interview Preparation
- HR Interview Preparation

- Mock Interviews
- Communication & Workplace Skills
- Job Role Guidance
- Recruitment Opportunity Updates
- Placement Assistance

Placement Information

Placement assistance may be provided to eligible learners based on available opportunities.

Placement is not guaranteed. Employer selection depends on candidate skills, qualification, performance, interview results, employer requirements, vacancies and other selection criteria.

COURSE LEVELS

FOUNDATION APP DEVELOPMENT

For learners beginning their programming and mobile development journey.

Programming Fundamentals → UI Basics → Mobile Application → Basic Project

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PROFESSIONAL APP DEVELOPMENT

For learners looking to develop practical mobile application skills.

Programming → Mobile UI → Application Logic → Database → API → Project

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ADVANCED APP DEVELOPMENT

For learners looking for more comprehensive application development exposure.

Advanced Programming → Mobile Framework → APIs → Authentication → Database → Testing → Deployment → Projects

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YOUR APP DEVELOPMENT JOURNEY

START

Computer & Programming Fundamentals

↓

LEARN

Programming Language & Development Tools

↓

DESIGN

Mobile UI & Application Screens

↓

DEVELOP

Build Functional Mobile Applications

↓

CONNECT

Database + APIs + Cloud Services

↓

TEST

Debug, Test & Improve Applications

↓

DEPLOY

Prepare Application for Release

↓

BUILD YOUR PORTFOLIO

Projects + Practical Experience + Career Preparation

WHY CHOOSE AAYUSH?

ESTABLISHED SINCE 2002

More than two decades of association with education and professional skill development.

PRACTICAL LEARNING

Focus on applying programming and development concepts through practical work.

MODERN TECHNOLOGY PATHWAYS

Android, Flutter and React Native pathways may be available depending on the selected programme.

PROJECT-BASED LEARNING

Learners can work on practical mobile application projects according to their level.

CAREER GUIDANCE

Career guidance is provided at no additional cost.

INTERNSHIP OPPORTUNITIES

Eligible learners may receive access to available internship opportunities, subject to organisation requirements and selection.

PLACEMENT SUPPORT

Career preparation and placement assistance are provided based on eligibility and available opportunities.

BUILD. TEST. LAUNCH.

Learn programming.
Build mobile applications.
Connect APIs and databases.
Work on practical projects.
Develop your portfolio.
Prepare for technology opportunities.

Start Your App Development Journey with Aayush Institute of Professional Studies.

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